

David Díaz-Villarejo
Ph.D. Candidate in Economics
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Education

Universidad Carlos III de Madrid Madrid, Spain
Ph.D. in Economics 2023 – present
Supervisors: Carlos Velasco Gómez, Jesús Gonzalo Muñoz

M.Res. in Economic Analysis 2023
B.Sc. in Economics 2021

Purdue University Indiana, U.S.
B.Sc. in Economics (Junior Year) 2019

Professional Experience

Universidad Carlos III de Madrid Madrid, Spain
Teaching Assistant (T.A.) 2022 – present

- Graduate Courses: Probability and Statistics Preliminary Course (Ph.D. in Economics; Fall, 2024-2026), Mathematics Preliminary Course (MSc. in Economic Development and Growth; Fall, 2024-2026).
- Undergraduate Courses: Econometrics (Spring, 2023-2026), Quantitative Microeconomics (Fall, 2025 & 2023), Time Series Econometrics (Fall, 2024), Microeconomic Theory (Fall, 2022), Mathematics for Economics II (Spring, 2022).
Teaching Excellence Awards in Econometrics and Quantitative Microeconomics.

UNIE Universidad Madrid, Spain
Lecturer. Fall, 2025

- Undergraduate Courses: Financial Mathematics (Business Administration, 2 groups)

Autoridad Independiente de Responsabilidad Fiscal (AIReF) Madrid, Spain
Research Assistant (R.A.) 10/2020 – 08/2021

- Long-term Forecasts and Sustainability Analysis of the Spanish Government Accounts for the Spanish Independent Fiscal Institution (IFI).

Analistas Financieros Internacionales, Afi (International Financial Analysts) Madrid, Spain
Professional Internship. 07/2019 – 12/2019

- Analysis of Spanish and European Credit Institutions for the Assets & Liabilities Committees (ALCOs).

Work in Progress

Gini Index Regression.
With Vedant Bhardwaj.

Proposes a parametric model for the conditional Gini index, in the spirit of mean and quantile regression. Derives the moment condition for a general parametric specification and develops econometric theory for

the linear case: identification results, including model misspecification, a loss function for the conditional Gini index, an R^2 -type goodness-of-fit measure, a Gini Index Regression (GIR) estimator, and bootstrap-based inference. An empirical application to U.S. labor market data reveals substantially higher inequality among non-union workers, showing that, among the college-educated, and controlling for characteristics, occupation, and industry, inequality is modest at labor market entry, consistent with the skill-biased technical change hypothesis, but rises sharply with experience.

Locally Robust Semiparametric Estimation of Parametric Conditional Gini.

With Vedant Bhardwaj, Juan Carlos Escanciano, and Joel R. Terschuur.

Derives the influence functions for functionals of conditional distributions to propose locally robust estimation and inference for Gini Index Regression (GIR) models, considering three alternative model specifications as leading examples, robust to first-stage, nonparametric kernel estimation of the conditional distribution function.

A Durbin-FGLS Approach to Robust Inference for Time Series Regression.

The paper proposes a feasible generalized least squares estimator based on time series Durbin regressions. The estimator is shown to be consistent under much weaker exogeneity assumptions than the standard OLS exogeneity, and allowing the number of first-stage lagged regressors to grow with the sample size. It is also shown to be more efficient than the existing FGLS variants proposed in the literature. Theoretical results are illustrated via Monte Carlo experiments.

Courses & Summer Schools

- Local Projection Methods for Time Series and Panel Data. CEMFI Summer School. Prof. Óscar Jordà (Federal Reserve Bank of San Francisco and UC Davis). 2024.
- DSGE and Time Series Models for Macroeconomic and Policy Analysis. CEMFI Summer School. Prof. Marco del Negro (Federal Reserve Bank of New York). 2024.
- Large Language Models and Machine Learning for Unstructured Data. Fundación Ramón Areces - IESE Business School. Prof. Stephen Hansen (UCL). 2024.
- Linear Programming and Economic Applications. Math+Econ+Code. Prof. Alfred Galichon (NYU). 2023.
- Applied Functional Analysis. Universidad Carlos III de Madrid. Prof. L. Alberto Ibort. 2023.

Honors & Awards

- Beca FPI para la Formación de Doctores. Ministerio de Ciencia e Innovación, 2023. Ph.D. Scholarship.
- Master's Scholarship. M.Res. in Economic Analysis (UC3M), 2021. Banco Santander. Full Scholarship.

Languages

Programming: R, Matlab, EViews, Stata, Excel.

Communication: English (C1), French (B2), Spanish (Native).

References

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